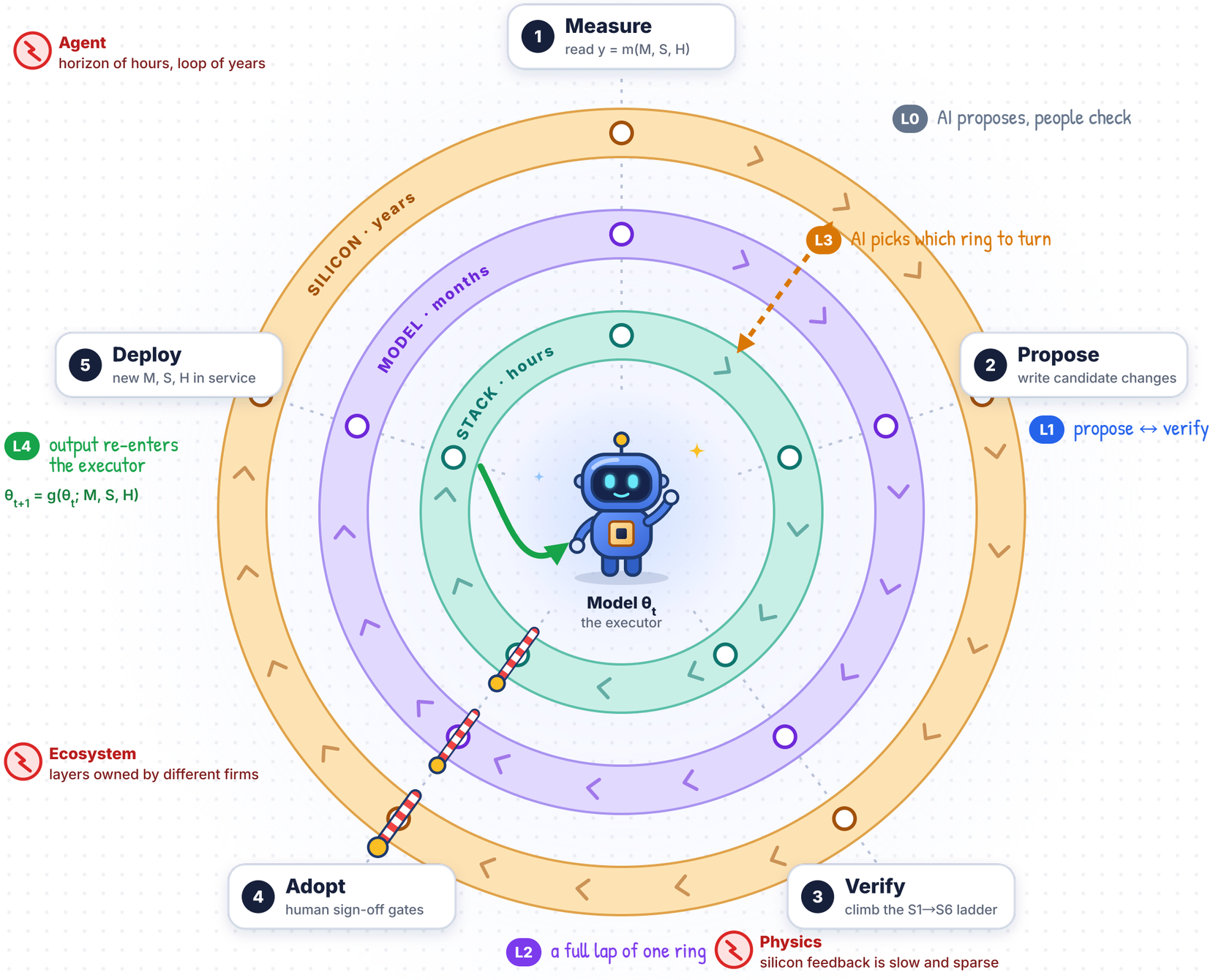
One turn of the co-evolution loop

three loops at three speeds share five stations; the model at the centre both turns them and is changed by them



WHAT HAPPENS AT EACH STATION

1 Measure

- **Stack** — profiles, end-to-end latency
- **Model** — evals, scaling-law fits
- **Silicon** — PPA reports, post-silicon data

reading across rings is what L3 needs

2 Propose

- **Stack** — kernels, schedules, serving configs
- **Model** — attention, numerics, MoE layout
- **Silicon** — RTL, HLS pragmas, μarch params

L0 suggests · L1 searches · L2 decomposes a spec

3 Verify

fewer candidates, slower, more trusted →

S1	S2	S3	S4	S5	S6
compile	function	PPA	profile	end-to-end	post-silicon
s	min–h	h	min	h–d	mo–yr

cheap tiers screen many; few reach silicon · reward hacking lives here

4 Adopt

- **Stack** — code review, merge
- **Model** — training-run approval
- **Silicon** — sign-off, tape-out decision

people hold the slowest gates

5 Deploy

- **Stack** — new kernels go live within hours
- **Model** — next model trained within months
- **Silicon** — new chips in service within years

when the new S or H runs the designer itself: L4